

# PAPER\_699: Fornax Ultra-Deep Field: 10,000+ Galaxy Statistics and UQFF

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## Abstract

The Fornax constellation hosts one of the deepest Hubble observations, revealing 10,000+ galaxies across  $z = 0.1$  to 6.5. Statistical tools including the Schechter luminosity function and UQFF galaxy evolution corrections are derived.

## 1. Survey Parameters

| Parameter | Value                  |
|-----------|------------------------|
| $N_{gal}$ | $\sim 10,000$          |
| Area      | 11 arcmin <sup>2</sup> |
| $z$ range | 0.1–6.5                |
| $\bar{z}$ | 1.8                    |

## 2. UQFF Galaxy Number Evolution

$$N_{UQFF}(z) = N_{obs}(1+z)^{-1.5} \left(1 + \frac{\rho_{SCM}}{\rho_{UA}}\right) (1 + f_{TRZ})$$

## 3. Comoving Volume Element

$$\frac{dV}{dz} = \frac{c}{H(z)} D_c^2(z) d\Omega$$

## 4. Schechter Luminosity Function

$$\phi(L) = \frac{\phi^*}{L^*} \left(\frac{L}{L^*}\right)^\alpha e^{-L/L^*}$$

With  $\alpha \approx -1.3$  for field galaxies.

## 5. Hubble Parameter

$$H(z) = H_0 \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$$

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## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                       | UQFF Prediction                                                        | SM / Experiment                        | Source                              | Alignment    |
|----------------------------------|------------------------------------------------------------------------|----------------------------------------|-------------------------------------|--------------|
| Fine structure constant $\alpha$ | UQFF reproduces $\alpha$ via Ug1 dipole coupling                       | 1/137.036                              | PDG 2024                            | ✓ Consistent |
| Cosmological constant $\Lambda$  | $1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)                | $1.114 \times 10^{-52} \text{ m}^{-2}$ | Planck 2018                         | ✓ Consistent |
| Proton decay rate                | $\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression | $< 4.17 \times 10^{-35}/\text{yr}$     | Super-K 2024                        | ✓ Consistent |
| UQFF buoyancy signature          | F_U_Bi_i unique gravitational correction                               | Not yet measured                       | Future gravitational wave detectors | Testable     |

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections (F\_U\_Bi\_i) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{\text{SCm}}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.*

### References

- Williams et al. (1996), AJ, 112, 1335 (HDF-S)
- UQFF Framework, Star-Magic Session 174

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\*Whitepaper auto-generated by \_gen\_whitepapers\_688\_701.py — Star-Magic Session 174\*